

Project Brief

Antenna Recommender — A Decision-Support Tool for Antenna Selection

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1. Purpose

The Antenna Recommender is a web-based application designed to help students and Communication engineers select the most suitable antenna for a given technical requirement. Users enter parameters such as frequency, application type, and required gain, and the system returns a recommended antenna, ranked alternatives, and a side-by-side comparison of key specifications.

2. Business Problem

Students and RF engineers often struggle to choose an appropriate antenna because selection depends on the interaction of several technical parameters (frequency, gain, application, beamwidth, and more). The resources available today are scattered across textbooks, datasheets, and vendor sites, and interpreting them correctly requires a level of specialist expertise that many users — particularly students and early-career engineers — do not yet have.

3. Proposed Solution

A responsive website that takes a user's technical inputs and runs them through a recommendation engine (decision-tree / lookup-table logic) validated by Subject Matter Experts. The tool returns one primary recommendation, at least one alternative, and a comparison table covering gain, beamwidth, polarization, size, and cost — condensing work that today requires specialist knowledge into a three-step guided flow.

4. Objectives & Success Criteria

Objective	Success Criteria
Scope completion	All planned activities — including project management documentation and the final web application — are completed and accepted by the Sponsor.
Schedule performance	Schedule Performance Index (SPI) ≥ 1.0 at project closure.
Functionality	The application recommends at least two distinct antenna types based on a user-defined frequency and application type.
Stakeholder satisfaction	Formal acceptance of the final deliverable by the Sponsor, reflecting complete requirement fulfillment.

5. Scope Summary

In Scope

- The Antenna Recommender website (front end and back end).
- The antenna recommendation algorithm.
- The user interface / user experience design.
- The antenna comparison feature.
- Full project management documentation.

Out of Scope

- Physical antenna testing or fabrication.
- Integration with real-time external databases.
- User accounts and login.

6. Target Users

- RF Engineers
- Communication Engineering students
- Researchers

7. Key Stakeholders

Stakeholder	Role
Dr. Randa Fouad	Project Sponsor — approves the project and provides overall guidance.
Nourin Elshenawi	Project Manager — plans, coordinates, and manages the project.
Project Team	Designs, develops, and tests the application.
Subject Matter Experts (SMEs)	Verify the technical accuracy of antenna recommendations.
End Users	Use the application and evaluate its usability.

8. High-Level Milestones

Milestone	Target
Initiation complete	Week 1
Planning complete	Week 5
Execution / testing complete	Week 6
Project closure	Week 7

9. Key Risks

- Technical data risk — difficulty sourcing reliable, validated datasets for the recommendation algorithm.
- Resource availability risk — potential scheduling conflicts among team members due to concurrent academic commitments.
- Scope creep — late requests for complex, specialized recommendation features not originally budgeted.